

Practice 1: (Textbook 1.12) Short put profit condition

A trader writes a December put option with a strike price of \$30. The price (premium) of the option is \$4. Under what circumstances does the trader make a gain?

TA Key for Practice 1

Writing (shorting) the put means the trader **receives** the premium \$4 now and may have to **pay** the put payoff at maturity. Let S_T be the stock price in December. Put payoff to the *holder* is $(30 - S_T)^+$, so the writer's profit is

$$(S_T) = 4 - (30 - S_T)^+.$$

Casework:

$$(S_T) = \begin{cases} 4, & S_T \geq 30, \\ 4 - (30 - S_T) = S_T - 26, & S_T < 30. \end{cases}$$

The trader makes a **gain** when $(S_T) > 0$, i.e. $S_T > 26$. Break-even at $S_T = 26$; losses occur when $S_T < 26$ (and the loss grows as S_T falls).

Practice 2: (Textbook 2.7) Margin call thresholds

A trader buys two July futures contracts on frozen orange juice concentrate. Each contract is for delivery of 15,000 pounds. The current futures price is 160 cents per pound, the initial margin is \$6,000 per contract, and the maintenance margin is \$4,500 per contract.

- What **price change** would lead to a margin call?
- Under what circumstances could \$2,000 be **withdrawn** from the margin account?

TA Key for Practice 2

Total contracts $N = 2$, contract size = 15,000 lb $\Rightarrow$ total underlying = 30,000 lb. Initial margin balance = $2 \times 6000 = 12000$. Maintenance level = $2 \times 4500 = 9000$.

(a) Margin call when balance *falls below* maintenance: loss $> 12000 - 9000 = 3000$. For a long futures position, loss occurs if price drops by $\Delta F < 0$:

$$\text{Loss} = (-\Delta F) \cdot 30,000.$$

Set Loss = 3000 gives $-\Delta F = 3000/30000 = 0.10$ dollars/lb = 10 cents/lb. So a **drop of more than 10 cents/lb** triggers a margin call (a 10-cent drop puts you right at maintenance).

(b) To withdraw \$2,000 and still meet initial margin, balance must be at least $12000 + 2000 = 14000$ before withdrawal. That requires profit of \$2,000:

$$\Delta F \cdot 30,000 = 2000 \Rightarrow \Delta F = 2000/30000 \approx 0.0667 \text{ dollars/lb} = 6.67 \text{ cents/lb}.$$

So if futures price rises by at least about **6.67 cents/lb**, the trader can withdraw \$2,000.

Practice 3: (Textbook 3.3) Optimal hedge ratio

Suppose that the standard deviation of quarterly changes in the **spot** price of a commodity is \$0.65, the standard deviation of quarterly changes in the **futures** price on the commodity is \$0.81, and the correlation between the two changes is 0.8.

- What is the **optimal hedge ratio** for a 3-month contract?
- What does it **mean** (interpretation in words / how you use it to pick # of contracts)?

TA Key for Practice 3

Optimal hedge ratio:

$$h^* = \rho \frac{\sigma_S}{\sigma_F} = 0.8 \cdot \frac{0.65}{0.81} \approx 0.8 \cdot 0.80247 \approx 0.642.$$

Interpretation: choose the futures position size to minimize the variance of the hedged P&L over the horizon. If hedging an exposure of Q units of the commodity with contract size q_F , then

$$N^* = h^* \frac{Q}{q_F},$$

taking **short futures** if you are long the spot (protect against price declines), and **long futures** if you are short the spot / need to buy later. Because $h^* < 1$, you hedge less than one-for-one due to imperfect correlation and differing volatilities.

Practice 4: (Textbook 4.18) Bond pricing and duration (continuous compounding)

A 5-year bond with a yield of 7% (continuously compounded) pays an 8% coupon at the end of each year. Assume face value is \$100 (so coupon is \$8 annually).

- (a) What is the bond's **price**?
- (b) What is the bond's **duration** (Macaulay duration, in years)?
- (c) Use duration to estimate the effect on the bond's price of a 0.2% **decrease** in yield.
- (d) Recalculate the bond's price using a 6.8% continuously compounded yield and verify agreement with (c).

TA Key for Practice 4

Cashflows: \$8 at $t = 1, 2, 3, 4$ and \$108 at $t = 5$. Yield $y = 0.07$ (cont.).

(a) Price:

$$P = \sum_{t=1}^4 8e^{-0.07t} + 108e^{-0.35} \approx 103.0513.$$

(b) Macaulay duration (continuous discounting):

$$D = \frac{1}{P} \sum_i t_i \text{PV}(CF_i) = \frac{1}{P} \left(\sum_{t=1}^4 t \cdot 8e^{-0.07t} + 5 \cdot 108e^{-0.35} \right) \approx 4.3235 \text{ years.}$$

(c) Small change approximation (cont. comp.): $\frac{\Delta P}{P} \approx -D \Delta y$. Here $\Delta y = -0.002$ so

$$\Delta P \approx P \cdot D \cdot 0.002 \approx 103.0513 \cdot 4.3235 \cdot 0.002 \approx 0.8911,$$

so P increases to about 103.94.

(d) Reprice at $y = 0.068$:

$$P(0.068) = \sum_{t=1}^4 8e^{-0.068t} + 108e^{-0.34} \approx 103.9466,$$

increase ≈ 0.8953 , close to duration estimate.

Practice 5: (Textbook 5.23) Index futures with dividend yield

An index is 1,200. The 3-month risk-free rate is 3% per annum and the dividend yield over the next 3 months is 1.2% per annum. The 6-month risk-free rate is 3.5% per annum and the dividend yield over the next 6 months is 1% per annum. All interest rates and dividend yields are continuously compounded. Estimate the futures price of the index for 3-month and 6-month contracts.

TA Key for Practice 5

Use $F_0 = S_0 e^{(r-q)T}$.

3-month ($T = 0.25$): $r = 0.03$, $q = 0.012$:

$$F_{0,3m} = 1200 e^{(0.03-0.012) \cdot 0.25} = 1200 e^{0.0045} \approx 1205.41.$$

6-month ($T = 0.5$): $r = 0.035$, $q = 0.01$:

$$F_{0,6m} = 1200 e^{(0.035-0.01) \cdot 0.5} = 1200 e^{0.0125} \approx 1215.09.$$